Long-Run Dynamics of Blockwise Self-Conditioned Language Generation After Prompt Eviction

Semantic Afterlife

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Abstract

A language model whose input is truncated to the last W generated tokens continues after the original prompt has left that window. We study the *blockwise re-prompt kernel* $Y_k^{(B)} \sim P_\theta(\cdot \mid X_k; B, \tau)$, $X_{k+1} = \mathrm{Tail}_W(X_k \oplus Y_k^{(B)})$, with B=1024 and W=4096 (B/W=0.25) on one instruct generator. After full seed eviction, a textual repetition lock is the typical T=0.3 outcome (19/20 domain trajectories). Last-band occupancy retains between-seed distinguishability on ten fixed seed texts (F4); that gap is not recovered prompt memory (H2). Hypothesis H1 (validated Markov state models) is unsupported. We did not measure a sliding native KV cache. Degeneracy is the sample, not a filter that was applied and then forgotten.

1 Introduction

Let $X_k \in V^W$ be the last W generator tokens. The object of this paper is not token-by-token sliding attention. It is the blockwise kernel

$$Y_k^{(B)} \sim P_\theta(\cdot \mid X_k; B, \tau), \qquad X_{k+1} = \mathrm{Tail}_W(X_k \oplus Y_k^{(B)}). \tag{1}$$

After generated count $g = W$ the seed is physically absent from the model's input (full eviction). The subsequent trajectory is the *afterlife*. The scientific question is whether that trajectory retains seed identity, occupies a domain-tagged lock, or diffuses.

That question is easy to answer wrongly. A looping trajectory occupies one point in representation space and reports a confined mean-squared displacement for reasons that have nothing to do with semantics. Degeneracy is labelled before geometry is read.

The experiments are Stages 0–6 of a prospectively specified, version-controlled sequential study with documented adaptive replanning between stages (`docs/research-plan.md`). They do not establish H1, H5, or a result for language models as a class.

Section 2 places the afterlife against verified related work. Section 3 defines the kernel, the two eviction times, and forced continuation. Section 4 reports the low-T lock. Section 5 reports last-band domain occupancy. Section 6 states what the design cannot establish. Section 7 lists remaining threats to validity.

2 Related work

We cite only sources marked VERIFIED in the project literature log.

Zekri et al. [2024] study autoregressive generation as a Markov chain on the context and, under contraction of the next-token kernel in total variation, prove existence of a unique stationary distribution. That theorem is the right null for a unique long-run law. It is not a measurement of last-band occupancy after eviction.

Telephone-game and paraphrase protocols pass the *whole* previous text under an instruction [Perez et al., 2025, Mohamed et al., 2025, Wang et al., 2025, Geng et al., 2026]. Our state is a finite tail with explicit prompt eviction and many turnovers of an imposed W. Ko and Geiping [2026] describe interlocutor-conditioned attractors; this protocol has no interlocutor. Chen et al. [2026] separate horizon, context, and memory; we impose W by truncation and begin after the seed has left that window. Wang et al. [2026] pin the prompt and randomly evict generated reasoning tokens for KV-cache compression; we evict the seed completely and continue uninstructed.

Surface-form degeneration under maximization decoding is documented [Holtzman et al., 2020]. Self-reinforcement of sentence repetition is the closest alternative explanation of the lock [Xu et al., 2022]. Training-time collapse on recursively generated data [Shumailov et al., 2024] is a different object: we never retrain. Markov-state methodology from molecular dynamics [Wu and Noé, 2020, Paul et al., 2019] is the toolkit Stage 3 applied; the toolkit is not the result.

3 Protocol

3.1 Generator, window, and P1

Occupancy uses `or-qwen3-8b` (Qwen3-8B-Instruct via OpenRouter; served provider `Alibaba` on every completed occupancy step), W=4096, T=0.3, block B=1024 so B/W=0.25, analysis chunks of 1024 generator tokens, non-overlapping, and twelve turnovers of W. The continuation mechanism is `raw_completion`, not `assistant_prefill` and not `chat_instructed`. Protocol P1 is the re-prompt: the last W tokens are re-encoded and sent as a new prompt. P1 and the continuation mechanism are distinct axes (glossary). The input construction is not a sliding native KV cache (protocol P2).

Eviction has two times. *Eviction start* is $W - L_0$: the window first fills and the next generated token can drop a seed token. *Full eviction* is $g = W$: no seed token remains. Last-band occupancy uses turnover ≥ 1, which is full eviction, not the fill instant. Historical generate JSONL stored `horizon_tokens` $= W - L_0$; analysis does not read that field for F4.

The process is *externally sustained self-conditioning after repeated termination attempts*: stop tokens are recorded and generation is continued. Survival analysis of EOS as absorbing is parked (Paper B).

3.2 Cost law

Hosted spend follows the recorded ledger. Stage 0 audit cost \$0.0128. Stage 1 generation billed \$6.32 on `s1-pilot-core-20260830T134147Z-f1cce9ec`. Stage 4 hosted generation billed \$3.4383. Stage 5 generation billed \$1.3385. Stage 6 used cached Gemini embeddings at hosted \$0. Project hosted spend at Stage 6 close was \$16.34 of a \$200 ceiling. This correctness pass spent \$0.

3.3 Degeneracy is the sample

A trajectory is labelled a textual repetition lock when calibrated repetition statistics cross a threshold derived from a human-prose reference corpus (Carroll and Darwin): mean 3-gram Jaccard 0.083 and late-window Jaccard 0.0122. The detector reports a near-recurrent textual

Table 1: Textual repetition-lock rates, Stage 2 model axis, split by temperature, W=4096, n=4 per cell, 95% Clopper–Pearson CI. Source: `artifacts/stage-2/model-axis/rates/fixed_point_rates_by_temperature.csv`. Pooled n=8 over $T \in \{0.3, 1.0\}$ remains in `fixed_point_rates.csv` and is not a T=0.3 table. Gemma 0/4 at both temperatures is the existence proof that the Stage 1 freeze is not a property of instruct models as a class.

Generator	T=0.3	95% CI	T=1.0	95% CI
`or-qwen3-8b`	3/4	$[0.194, 0.994]$	4/4	$[0.398, 1]$
`or-gemma-4-31b`	0/4	$[0, 0.602]$	0/4	$[0, 0.602]$
`or-gpt-oss-120b`	4/4	$[0.398, 1]$	4/4	$[0.398, 1]$
`or-muse-glimmer-30b`	2/4	$[0.068, 0.932]$	0/4	$[0, 0.602]$
`or-gpt-oss-20b`	did not complete design T			

Table 2: Looping rate by temperature and window, Stage 4, `or-qwen3-8b`, n=4 per cell, Clopper–Pearson CI. Source: `artifacts/stage-4/grid/looping_rate_by_cell.csv`.

T	W=4096	CI	W=8192	CI
≤ 1.0	4/4	$[0.398, 1]$	4/4	$[0.398, 1]$
1.5	0/4	$[0, 0.602]$	2/4	$[0.068, 0.932]$

regime (fixed point or short cycle), not an exact period-1 test. Those two numbers are not retuned.

On the occupancy panel, 19/20 domain trajectories and 1/2 love trajectories meet the lock rule; 10/10 domain seeds have at least one lock. Clopper–Pearson 19/20 is $[0.751, 0.999]$. Occupancy is a statement about locked trajectories. Geometry on a looping row is not interpreted as a confined semantic basin.

4 Low-temperature repetition lock

Stage 1 generated 48 trajectories on `or-qwen3-8b` at W=4096, target T=131072 tokens (32 turnovers). The lock rule fired on 45/48 trajectories (94%), including 24/24 at T=0.3 and 21/24 at T=1.0. Figure 1 shows a domain-gap curve on that degenerate-majority panel; it is not a non-looping subsample.

The lock is not universal across hosted generators. Table 1 splits Stage 2 rates by temperature, W=4096, n=4 per cell, Clopper–Pearson. The previously published n=8 rows pool $T \in \{0.3, 1.0\}$: `or-gemma-4-31b` 0/8 ($[0, 0.369]$), `or-gpt-oss-120b` 8/8 ($[0.631, 1]$), `or-qwen3-8b` 7/8 ($[0.473, 0.997]$). At T=0.3: gemma 0/4 ($[0, 0.602]$), 120B 4/4 ($[0.398, 1]$), Qwen 3/4 ($[0.194, 0.994]$). At T=1.0: gemma 0/4, 120B 4/4, Qwen 4/4. Direction (F2) is gemma versus 120B at both temperatures. Qwen's T=0.3 interval includes 0.5.

A prefill-versus-raw arm on `or-qwen3-8b` is 7/8 versus 8/8; Newcombe difference CI includes 0 (Fisher p=1).

Stage 4 crossed temperature and window with n=4 per cell. At $T \leq 1.0$, both W=4096 and W=8192 are 4/4 ($[0.398, 1]$). At T=1.5, W=4096 is 0/4 ($[0, 0.602]$); W=8192 is 2/4 ($[0.068, 0.932]$). That 4/4 versus 0/4 cell contrast at W=4096 is not a temperature law. Hypothesis H5 (a temperature window of non-looping confined motion) is absent: clean MSD α is defined only at T=1.5 and is subdiffusive.

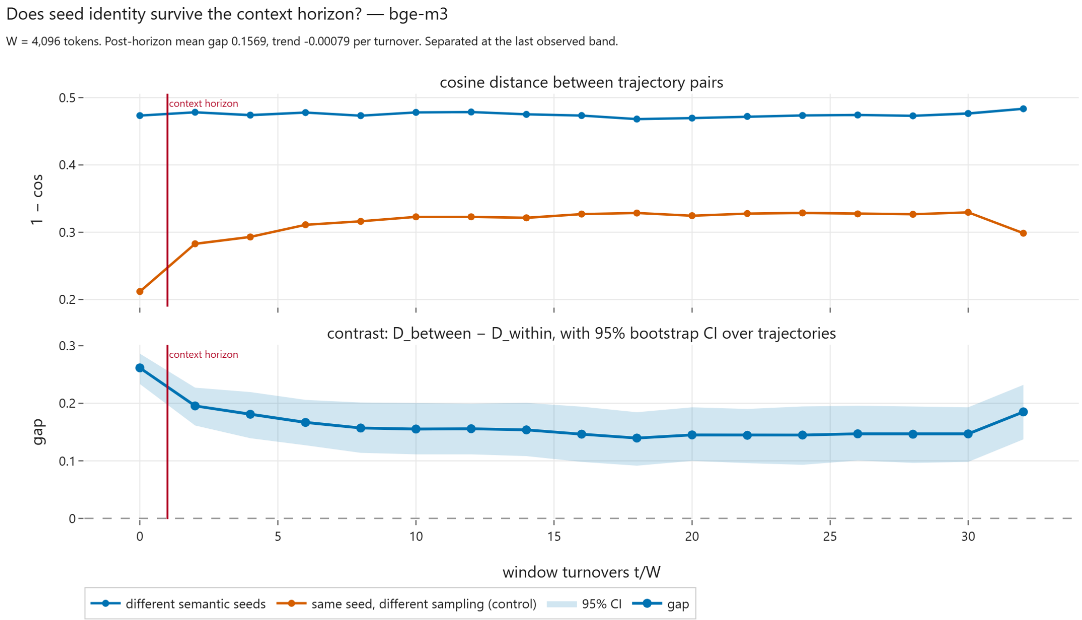

Figure 1: Stage 1 domain-gap versus turnover in `bge-m3` on the 48-trajectory `or-qwen3-8b` pilot. Degenerate-majority panel. Data: `artifacts/stage-1/separation-bge-m3/seed_separation.data.parquet`.

5 Last-band occupancy

Given the lock, Stage 5–6 ask whether last-band centroids of ten fixed seed texts remain distinguishable (between-seed vs within-seed). Twenty-eight trajectories at W=4096, T=0.3, B=1024, twelve turnovers. Generate `s5-lock-occupancy-20260905T164327Z-6780902f`; embed `s5-embed-lock-occupancy-20260906T030125Z-eab6e484`; degeneracy `s5-degeneracy-20260906T030145` gemini `s6-embed-third-space-20260906T082301Z-588eff8f`.

Let D_{within} be the mean cosine distance among last-band centroids that share a seed text, and D_{between} the mean among different seed texts. The gap is $\Delta_{\text{dom}} = D_{\text{between}} - D_{\text{within}}$. Domain here means one labelled seed instance, not a sampled population. Seeds are labelled separated when the trajectory-bootstrap 95% CI excludes 0 from above. That estimator already resamples with multiplicity (ADR-0019). An independent seed-pair matrix recompute agrees on the point to < 0.002 in all three spaces. Those published CIs are not re-derived from embeddings: named occupancy run directories are not recoverable (ADR-0021). Regenerating trajectories would mint new `run_ids` and new draws, not verify the published intervals.

Table 3 reports last-band F4. All three spaces are separated under that CI rule: `bge-m3` 0.201 $[0.065, 0.332]$; `qwen3-embed-8b` 0.390 $[0.151, 0.593]$; `gemini-embed-001` 0.150 $[0.029, 0.266]$. Sign agreement is the claim. Levels are not interchangeable. Gemini's lower bound 0.029 is an NHST pass, not architecture-independence. Physics seed 1 has 47 versus 48 chunks, so its last-band D_{within} diagonal is undefined; F4 uses n_{within}=9 pairs. Leave-one-seed-out on the matrix keeps the gap positive in every space (minimum 0.128). Within-pair randomisation of the 54 unique finite pairs yields p=0.0001 in each space (0/9999 permutations as large as the observed gap, seed 0). That is a test on ten fixed seed texts, not a domain population. The last-band gap is between-seed distinguishability, not recovered prompt memory (H2) and not domain memory.

A protocol diagnostic, not a headline: domain trajectories spend a mean fill fraction 0.853, 95% CI $[0.756, 0.936]$, in the last quarter, with mean stop-token fraction 0.359.

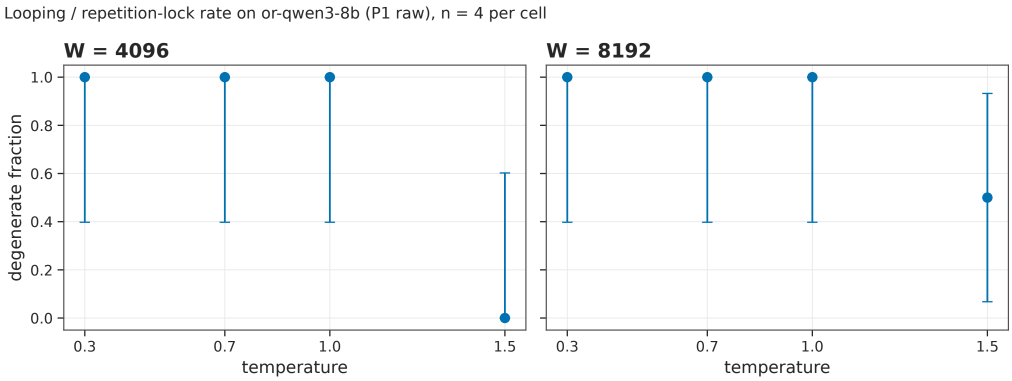

Figure 2: Looping rate versus temperature at two windows, 95% Clopper–Pearson CI, n=4. The T=1.5, W=4096 cell is the existence of a non-looping regime on this model, not a mixing time.

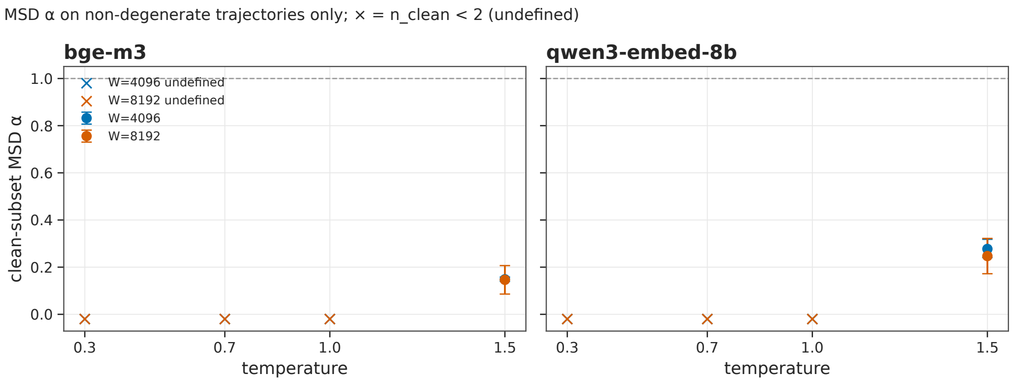

Figure 3: Clean MSD α where defined (non-looping cells only). Missing low-T points are the H5 absence.

6 What this does not show

H1 is unsupported. Stage 3 applied VAMP/PCCA+ on embeddings of the Stage 1 panel. `validated=0` in every cell of `artifacts/stage-3/dynamics/k_stability.md`. Implied n_{macro} is unstable across K and across embedding spaces. Hypothesis H1 (metastable semantic states as an MSM object) is unsupported. A short-cycle lock is not a validated macrostate.

H5 is absent. There is no low-T clean α on the Stage 4 grid.

F6 is not a headline. Matched replica pairs are counterfactual narratives, not one-token factual flips. n=2 per family. Published twin CI columns used a bootstrap that dropped resample multiplicities (ADR-0019) and are not reported here as intervals. Last-band *point* Δ in `gemini-embed-001` is -0.012 (all pairs) and 0.049 in `bge-m3` waterloo. That is no detected divergence / underpowered, not occupancy of one lock and not semantic collapse. An equivalence design is not claimed.

No universal attractors. One instruct generator, P1, `raw_completion`, Alibaba, T=0.3. Gemma's 0/8 already falsifies universality.

Table 3: F4 last-band between-seed vs within-seed gap, ten fixed seed texts, n=20, n_{within}=9. Source: `artifacts/stage-6/occupancy/domain_separation_last_band.csv`. Separated means the published 95% CI excludes 0 from above. Published 95% CIs are original analysis outputs in that CSV; source occupancy vectors are not recoverable (ADR-0021).

Space	gap	95% CI	separated
`bge-m3`	0.201	$[0.065, 0.332]$	yes
`qwen3-embed-8b`	0.390	$[0.151, 0.593]$	yes
`gemini-embed-001`	0.150	$[0.029, 0.266]$	yes

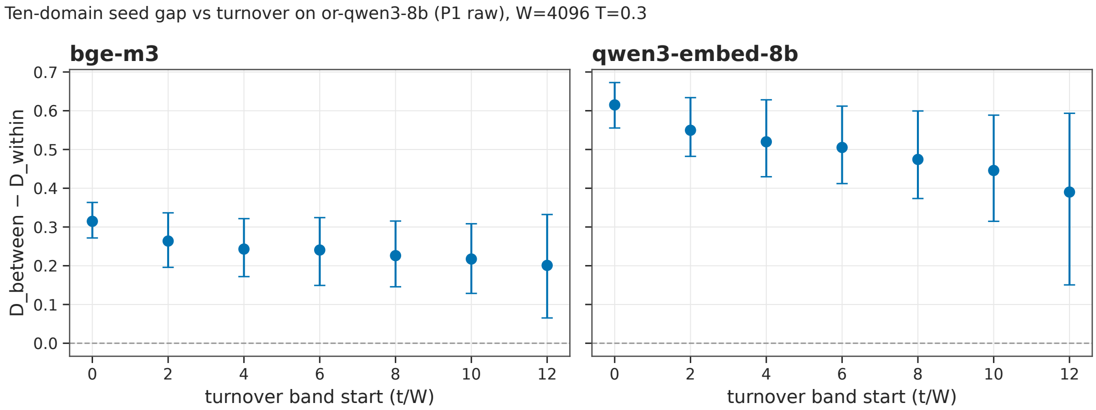

Figure 4: Between-seed last-band gap versus turnover in `bge-m3` and `qwen3-embed-8b` on the Stage 5 locked panel. Last-band CIs exclude 0 from above in both spaces.

7 Limitations

P1 versus sliding attention. Every afterlife token is produced by re-prompting the last W tokens. A native sliding KV cache (P2) was not measured. No sentence here is evidence about sliding attention.

One generator, one protocol, one temperature for occupancy. F4 is `or-qwen3-8b`, P1, `raw_completion`, served provider `Alibaba`, T=0.3, W=4096, B=1024. P1 is the re-prompt axis; `raw_completion` is the continuation mechanism. They are not synonyms.

Instruction-tuning and register. The occupancy generator is instruct-tuned. Stage 2 already measured a reviewer register on this model. A persona/register alternative — last-band centroids cluster by residual instruction style rather than seed domain — is not excluded by F4. No hosted base model was available (Stage 0). A local `gemma-3-1b-pt` probe at W=256 is a different window.

Forced continuation. The object is externally sustained self-conditioning after repeated termination attempts. Last-quarter stop fraction 0.359 is a confound to name, not an EOS-absorbing measurement.

Twins, n=2. F6 is underpowered. Twins are not one-fact pairs. CIs from the broken bootstrap are not used as evidence.

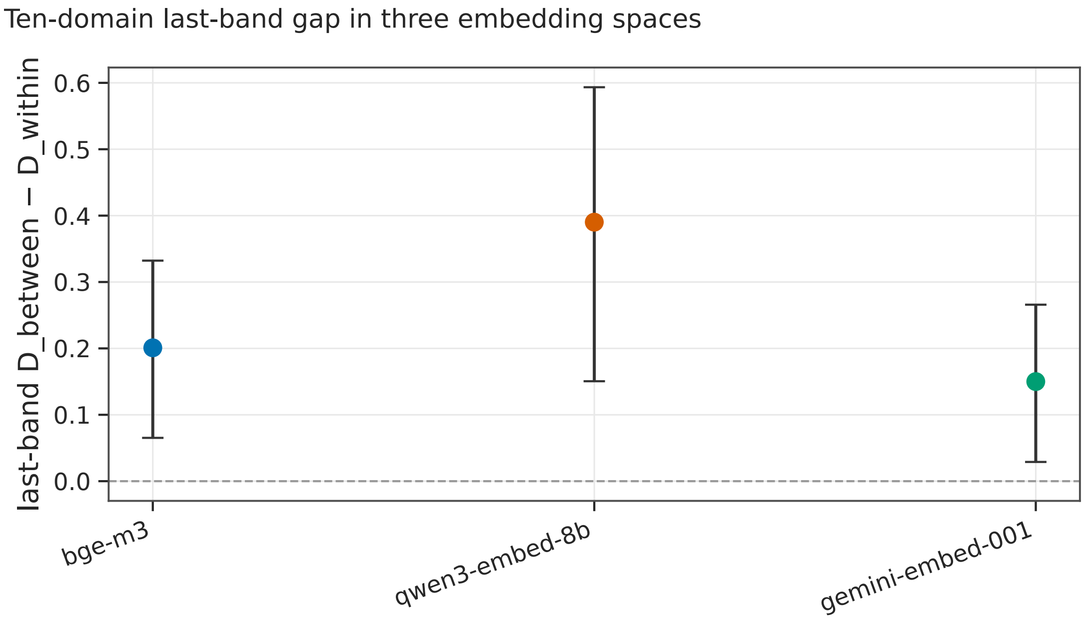

Figure 5: F4 between-seed last-band gap versus turnover in three embedding spaces on the same 28 trajectories. Last-band CIs exclude 0; gemini lower bound is 0.029. Levels 0.201 / 0.390 / 0.150 are not interchangeable.

Degeneracy null. The lock bar is Carroll+Darwin prose. Wikipedia-scale calibration is parked.

Representation. Absolute gaps 0.201, 0.390, 0.150 disagree. Gemini lower CI 0.029 is not architecture-independence.

Provider non-determinism. Stage 2 exact-match rates were 100/60/20/20 percent across four hosted generators. Replica pairs are not guaranteed independent draws.

Occupancy vectors. Named occupancy generate and embed run directories are not recoverable (ADR-0021). Published trajectory-bootstrap CIs are the original analysis outputs in the committed CSV, not re-derived from embeddings. Independently reproducible F4 inference in this pass is the seed-pair matrix, leave-one-seed-out, and within-pair randomisation. A new generate would be a new panel, not a restore of those intervals.

Finite horizon. Occupancy uses twelve turnovers, not an infinite afterlife.

8 Discussion

What is established, on this generator and protocol: after the seed leaves W, the typical T=0.3 trajectory is a textual repetition lock; last-band centroids still show a between-seed last-band gap across ten fixed seed texts in three embedding spaces (not recovered prompt memory, H2).

What is not established: H1; H5; we cannot claim architecture-independence of occupancy; a result for language models as a class; P1 versus P2; mixing times; replica collapse; one-fact twin sensitivity.

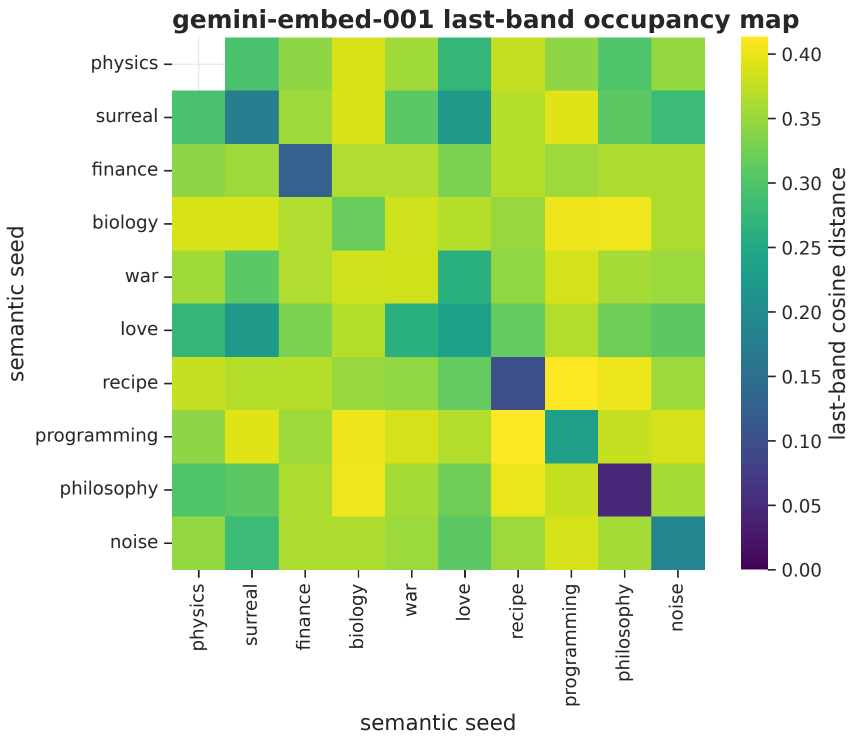

Figure 6: Last-band cosine distances in `gemini-embed-001` (illustration of the pairwise matrix, not a cluster count, not a 2-D projection).

Zekri et al. [2024] remain the right uniqueness null. Seed-tagged last-band gaps on a locked panel are compatible with more than one long-run occupancy region in embedding space. That compatibility is not a proof that the token-level chain has several stationary measures.

9 Reproducibility

Experiments through Stage 6 closed on git commit `ad1f244`. Stage 7 wrote the first manuscript. This correctness pass is ADR-0019, ADR-0020, and ADR-0021 on branch `cursor/tmlr-correctness-a7f7`; it does not mint `runs/s7`. Hosted spend for the pass is $0. The GitHub tag `state-latest` is a mutable restore pointer, not a citable archival object.

Primary `run_ids`:

- Stage 1 generate: `s1-pilot-core-20260830T134147Z-f1cce9ec`
- Stage 1 degeneracy: `s1-degeneracy-20260831T070038Z-b2f6f0c9`
- Stage 1 embed: `s1-embed-pilot-core-20260831T064430Z-f05e6a68`
- Stage 1 separation: `s1-separation-bge-m3-20260831T074042Z-9bf54fbf`
- Stage 2 model axis: `s2-model-axis-20260901T015457Z-ab59afc8`
- Stage 2 mechanism: `s2-mechanism-20260901T071519Z-dfbb173a`
- Stage 2 rates: `s2-rates-20260901T125506Z-41d2e88e`
- Stage 3 dynamics: `s3-dynamics-20260901T204521Z-f21d5908`

- Stage 4: `s4-w4096-new-temps-20260904T103121Z-589c8eb1`, `s4-w8192-20260904T120057Z-ce82ce55`
- Stage 5 generate: `s5-lock-occupancy-20260905T164327Z-6780902f`
- Stage 6 gemini: `s6-embed-third-space-20260906T082301Z-588eff8f`

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